

PROJECT PROPOSAL BGI NEXUS HACKATHON

Project Name: S.A.N.T.A.

(System for Authenticating Neighborhood Trust and Accountability)

Powered by: OmegaClaw AI & HandyHouseHelp (H3)

Author: Kristian Dansen

kristiandansen@gmail.com

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Mission:

Most of the AI field is racing to scale intelligence first and worry about values later.

S.A.N.T.A. flips this model by embedding compassion, community, and verifiable integrity into the AI architecture from the first line of code. Our mission is to support localized trust in the home service industry by establishing a verifiable, non-extractive index of professional integrity.

1. The Problem:

The Erosion of Digital Trust and Information Asymmetry

The digital marketplace for local home services is increasingly fragmented. Modern search directories and lead-generation platforms often fail to connect consumers with honest, verified professionals. Instead, they frequently favor entities optimized for ad spend, high-budget marketing, and search-engine manipulation.

As a result, independent, highly skilled local artisans are often marginalized, while consumers struggle to distinguish reputable providers from low-quality operators or predatory networks. The industry lacks a transparent, open, and verifiable infrastructure capable of evaluating local professionals based on objective, community-grounded trust metrics rather than marketing capital.

2. The Proposed Solution:

Project S.A.N.T.A.

Project S.A.N.T.A. is proposed as an AI-assisted "Integrity Engine" designed to reconstruct the digital trust layer for local services. Operating on a collaborative human-AI architecture, the system systematically maps, crawls, and verifies the reputational standing and operational compliance of service providers within specific geographic regions.

Rather than relying on self-reported reviews or commercial advertising data, S.A.N.T.A. evaluates businesses against objective regional standards, trade credentials, public registries, and local community contributions.

3. How the S.A.N.T.A. Core Architecture Works

To balance scalability with computational and factual reliability, S.A.N.T.A. utilizes a structured, five-stage pipeline:

Phase 1: Geographic & Semantic Mapping

Because the criteria for operational "quality" and legitimacy vary across jurisdictions, the system divides target markets into regional clusters. The primary AI agent (OmegaClaw) first researches the specific regulatory and community-trust landmarks of that area—identifying required local licenses,

active trade associations, mandatory municipal registries, and localized indicators of community commitment (such as local sports sponsorships or chamber of commerce memberships).

Phase 2: Autonomous Harvesting & Expert-in-the-Loop Validation

The AI agent crawls structured and unstructured primary sources (municipal databases, business registries, and trade directories). To mitigate hallucination risks and address accessibility barriers (such as CAPTCHAs, paywalled records, or nuanced local regulatory phrasing), the pipeline integrates a structured Human-in-the-Loop escalation protocol. When encountering high-uncertainty data, the AI routes the query to local community operators for micro-verification.

Phase 3: Regional Saturation

The AI dynamically aggregates and cross-references data points until a specific postal area or municipality reaches a statistically representative saturation point of validated local service providers. Once this threshold is achieved, the system generates a localized index.

Phase 4: Proactive Onboarding

S.A.N.T.A. executes proactive, automated outreach (via structured email or voice agents) to the identified high-integrity professionals. Providers are invited to review their verified credentials, sign an open pledge of professional honesty, and claim their profile on the public index at no cost. S.A.N.T.A. will share this list publicly for all to find.

Phase 5: HandyHouseHelp Integration

Once onboarded, these verified professionals are integrated into the HandyHouseHelp (H3) ecosystem. This provides them with an open, non-extractive channel to be discovered by local homeowners. The platform will link to good local companies for free. And will later offer a complete track and trace from quotation request to delivered job. This will log all context from each phase for evaluation if something goes wrong.

4. The Open Trust Community & Local Context Adapters

S.A.N.T.A. is designed as an active, sociotechnical network rather than a static directory. Customer performance data, project outcomes, and local compliance updates are continuously ingested. The system incorporates local context adapters to parse behavioral nuances, recognizing that professional expectations, communication standards, and regulatory compliance paths differ significantly across different regional cultures (e.g., comparing urban European standards to rural North American settings).

5. Our Goal at the BGI Nexus Hackathon

During this 3-day production sprint, our team is competing in Track 2. Our objective is to transition S.A.N.T.A. from a concept into a concrete, reviewable artifact within the ASI: Create ecosystem. By Sunday, we will deliver:

1. **An OmegaClaw Agent Workflow in ASI:** Create: A prototype workflow demonstrating how OmegaClaw autonomously crawls and cross-references localized "Sources of Truth" (e.g., municipal registries) to verify a home service provider.
2. **The Human-in-the-Loop Escalation Demo:** A guided demo/walkthrough illustrating our routing logic. It will show how OmegaClaw handles high-friction, ambiguous data by requesting micro-verification from a human community validator, proving how human compassion and local context work alongside the AI.
3. **A Reusable Task Template:** A blueprint that the broader SNET community can use to adapt our localized trust-verification workflow for other industries (e.g., automotive repair, agriculture).

6. Beyond the Hackathon: Next Steps & The "Law as Code" Integration

While the hackathon focus is on the discovery and validation of trusted professionals, the long-term roadmap for S.A.N.T.A. and HandyHouseHelp involves addressing the downstream lifecycle of service transactions—specifically, pre-contractual alignment, immutable execution tracking, and dispute resolution.

To build a structurally trustworthy framework, we propose extending Jörg Resch's "Law as Code" architecture (historically utilized for public administrative law) into the private, regulated home-services economy. This converts S.A.N.T.A. from a directory into an end-to-end normative infrastructure.

POST-HACKATHON ROADMAP



Phase 6: Dispute Prevention via Fact Schemas and Decision Trees

Disputes typically originate from misaligned pre-contractual expectations. To prevent this:

- Fact Schemas: Homeowners document their property conditions using formally typed schemas (e.g., building age, current equipment models, structural materials, and existing performance scores).
- Decision Trees: S.A.N.T.A.'s rule engines translate technical building norms

(such as NEN 2767 standards), manufacturer specifications, and municipal building codes into executable logical paths.

- The Decision Package: Before any physical work begins, the system generates a structured Decision Package detailing the exact scope of compliant work, required qualifications, and necessary steps. This eliminates information asymmetry between the contractor and the homeowner from the outset.

Phase 7: Immutable Progress Verification via the Registry

To ensure accountability during execution, we transition from verbal assertions to verifiable evidence:

- Both the homeowner and the service provider contribute step-by-step visual documentation (images, sensor data, or diagnostic readouts) at predefined milestones of the project.
- This documentation is written to a federated, append-only, and cryptographically signed Registry. This creates a tamper-proof record of what was built, which norms were followed, and what materials were used, anchored directly to the homeowner's digital property dossier.

Phase 8: Post-Execution Mediation & The Living Regulatory Library

When a sub-optimal outcome or dispute occurs, the system utilizes a collaborative human-AI resolution protocol:

- The Living Regulatory Library: OmegaClaw maintains a structured, continually updated library of building codes, product manuals, and technical specifications.
- Coverage Maps: The dispute is parsed against a Coverage Map to distinguish between strict, formalized technical standards (where compliance is binary) and areas requiring subjective, professional human judgment.
- The community and the AI analyze the immutable project timeline (the Registry) and the pre-contractual blueprint (the Decision Package) to reach an objective, transparent resolution.

Phase 9: Continuous Learning Loops and Multi-Vertical Scaling

The resolution process does not operate in a vacuum. The specific context, technical failure points, and behavioral nuances of each resolved dispute are analyzed to extract generalized lessons. This metadata is fed back into the pre-contractual Fact Schemas and Decision Trees, refining future quotation and scope-of-work engines to prevent similar errors.

While initiated within the real estate maintenance and home renovation sector, this standardized compliance-and-trust architecture is designed to scale across other highly fragmented, highly regulated private verticals (such as local automotive repair, specialized agriculture services, and localized commercial logistics).

7. Strategic Fit within the BGI Nexus Ecosystem

Progress on BGI depends on whether the wider community can build with, extend, and apply these systems to messy, real-world problems. S.A.N.T.A. is designed as a bridge connecting BGI's theoretical architecture with tangible, localized human interaction. It serves as a real-world grounded layer that integrates systematically with:

- OmegaClaw (Decentralized Memory & Investigation): Acts as the primary agent for localized data crawling, maintaining historical compliance records, and parsing complex manufacturer and regulatory documents.
- Utsa/OfferNet (Coordination & Matching): Utilizes S.A.N.T.A.'s verified registry to enable precise, regulation-aware matching of local needs with certified professional capacities.
- Sensemaking (Value Harvesting & Trust Assessment): Captures community-driven qualitative feedback and regional standards of trust, turning subjective local sentiment into structured reputational markers.
- Sage Lumina (Situational & Safety Awareness): Evaluates the safety, environmental, and human-impact implications of specific regulatory choices or compliance failures during project scoping and dispute mediation.

8. Our Ambition & The Road Ahead

We are building S.A.N.T.A. out in the open to help shape the kind of intelligence we want in our world—intelligence that protects and uplifts communities.

Our immediate goal for this sprint is to be selected as a track finalist. We are seeking the opportunity to present our artifact live to Ben Goertzel, the SNET development teams, and BGI ecosystem leaders. We want to gather high-level architectural feedback on our Human-AI validation loop and explore strategic pathways to make S.A.N.T.A. a native, open-source primitive for transaction trust across the entire SingularityNET ecosystem.